

MFAI (July-Dec/26) - Quiz-1-Solutions

01/September/2026 ;

Answer (3) **and** either (1) or (2) : Each carries 5 marks ; Time - 60 minutes.

1. For $\lambda \in \mathcal{R}$, define $A_\lambda = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 2 & 5 & -7 & -5 \\ 2 & -1 & 1 & 3 \\ 5 & 2 & -4 & \lambda \end{bmatrix}$. Define $b = \begin{bmatrix} 1 \\ -2 \\ 4 \\ 6 \end{bmatrix}$. Use

Gaussian Elimination to classify all values of λ according to whether $A_\lambda x = b$ admits no solution, unique solution, finitely many solutions or infinitely many solutions. Specify each elementary row operation involved.

Solution : Applying row-elementary operations $R2 \leftarrow R2 - 2R1, R3 \leftarrow R3 - 2R1, R4 \leftarrow R4 - 5R1$ to $[A_\lambda, b]$, we obtain $[B_\lambda, b_1] := \begin{bmatrix} 1 & 1 & -1 & -1 & 1 \\ 0 & 3 & -5 & -3 & -4 \\ 0 & -3 & 3 & 5 & 2 \\ 0 & -3 & 1 & 5 + \lambda & 1 \end{bmatrix}$.
Apply $R3 \leftarrow R3 + R2$ and $R4 \leftarrow R4 + R2$ to get $[C_\lambda, b_2] := \begin{bmatrix} 1 & 1 & -1 & -1 & 1 \\ 0 & 3 & -5 & -3 & -4 \\ 0 & 0 & -2 & 2 & -2 \\ 0 & 0 & -4 & 2 + \lambda & -3 \end{bmatrix}$.
Apply $R4 \leftarrow R4 - 2R3$ to get $[D_\lambda, b_3] := \begin{bmatrix} 1 & 1 & -1 & -1 & 1 \\ 0 & 3 & -5 & -3 & -4 \\ 0 & 0 & -2 & 2 & -2 \\ 0 & 0 & 0 & \lambda - 2 & 1 \end{bmatrix}$.

Hence, if $\lambda \neq 2$, we get the unique solution

$$x_4 = \frac{1}{\lambda - 2}, x_3 = 1 + x_4 = \frac{\lambda - 1}{\lambda - 2}, x_2 = \frac{-4 + 5x_3 + 3x_4}{3} = \frac{\lambda + 6}{3\lambda - 6},$$

$$x_1 = 1 - x_2 + x_3 + x_4 = \frac{5\lambda - 12}{3\lambda - 6}.$$

If $\lambda = 2$, then $A_\lambda x = b$ admits no solution.

2. Apply Gramm-Schmidt Orthogonalisation process to the basis $C := [c_1, c_2, c_3]$, where

$$c_1 := \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, c_2 := \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, c_3 := \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

to obtain an orthonormal basis of R^3 . Use $x^T y$ as the inner product.

Solution : Since $\|c_1\|_2 = 1$, we have $b_1 = c_1 = e_1$. Define $z_2 := c_2 - (c_2^T b_1) b_1 = c_2 - b_1 = e_2$ and hence define $b_2 = e_2$. Define $z_3 = c_3 - (c_3^T b_1) b_1 - (c_3^T b_2) b_2 = c_3 - b_1 - b_2 = e_3$. Hence, define $b_3 = e_3$. Thus, the orthonormal basis obtained is $[e_1, e_2, e_3]$.

3. Consider the linear mapping $\phi : R^3 \rightarrow R^3$ defined by

$$\phi(e_1) = b_1 := \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \phi(e_2) = b_2 := \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}, \phi(e_3) = b_3 := \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix},$$

where $I_3 := [e_1, e_2, e_3]$ forms the standard basis and $B := [b_1, b_2, b_3]$.

- Justify (with brief arguments) that ϕ is invertible.
- Determine the matrix X representing ϕ with respect to basis I_3 for both the domain and co-domain. Similarly, determine Y representing ϕ with respect to basis B for both domain and codomain.
- Determine $P \in R^{3 \times 3}$ satisfying $X = PYP^{-1}$. P should be determined logically and not by guess.
- Define $S := \{(x, y, z) \in R^3 : x^2 + y^2 + z^2 \leq 1\}$. Determine $\phi(S)$.
- Determine $\phi^{-1}(S)$.

Solution : (a) : For any $u = (x, y, z)^T \in R^3$, $\phi(u) = xb_1 + yb_2 + zb_3 = (3z, x, 2y)^T$. Hence, $\phi(u) = 0 \in R^3$ if and only if $u = 0 \in R^3$. Hence, ϕ is

invertible. (b) : By definition, $X = \begin{bmatrix} 0 & 0 & 3 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$. Since $b_3 = 3e_1, b_2 = 2e_3$ and $b_1 = e_2$, we have $\phi(b_1) = b_2, \phi(b_2) = 2b_3$ and $\phi(b_3) = 3b_1$. Hence, $Y = \begin{bmatrix} 0 & 0 & 3 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$.

(c) : Q denotes the matrix mapping coefficient vectors of I_3 to coefficient vectors with respect to B_3 . Q^{-1} is the matrix mapping coefficients with respect to B to those with respect to I_3 . By definition, $Q = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1/2 \\ 1/3 & 0 & 0 \end{bmatrix}$

and $Q^{-1} = \begin{bmatrix} 0 & 0 & 3 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$. It can be verified that $QQ^{-1} = I_3 = Q^{-1}Q$. Also,

it can be verified that $X = Q^{-1}YQ$. Renaming Q^{-1} as P and Q as P^{-1} , we obtain the desired conclusion.

(d) and (e) : We have

$$\phi(S) = \left\{ \begin{bmatrix} 3z \\ x \\ 2y \end{bmatrix} : x^2 + y^2 + z^2 \leq 1 \right\} = \left\{ \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} : \frac{\alpha^2}{9} + \beta^2 + \frac{\gamma^2}{4} \leq 1 \right\}.$$

$$\phi^{-1}(S) = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x^2 + 4y^2 + 9z^2 \leq 1 \right\}.$$